

Quality Control On-line System

Thickness Gauge
Surface inspection
Web Cleaner
Pinning

Leader in the WEB Quality
Control System

고객은 기업활동의 일선에 있습니다.
그들의 만족과 성공은 저희 성공의 기초입니다.
개발에서 서비스에 이르기까지 모든 영역에서 고객의
욕구 및 요구사항에 가능한 최선을 다하는것이
저희의 목표입니다.
저희는 고객이 보다 합리적이고 정확하며 편하게
목표를 달성하도록 최선을 다하겠습니다.

MEK
SYSTEM TECHNOLOGY

www.mekeng.com



About MEK Engineering Corporation. / MEK Inc.

Quality Control On-Line System

Thickness Gauge, Web Cleaner,
Pinning System

MEK는 움직이는 WEB품질관리 시스템의 선도적인 기술 전문 기업입니다.

지난 30년간, MEK는 지속적인 연구 개발을 바탕으로 풍부한 경험과 기술력을 쌓으며,
Roll to Roll 공정에 대한 세계 최고 수준의 계측 및 품질관련 핵심 시스템을 공급해 왔습니다.
IT와 포장용에 이어 2차전지, Display 등의 전자 부분과 Copper foil, Al foil, Glass, Steel 등
각종 첨단 소재 분야에서도 최고, 최적의 해결책으로 고객으로부터 신뢰받는 품질 관리 시스템을
세계 26개국에 공급하고 있습니다.

• 대표 제품: Online thickness gauge, Surface inspection, Web cleaner, Pinning system

MEK is the company specializing in the WEB quality control system.

For last 30 years, MEK have supplied high class thickness gauging system and other irreplaceable systems
for the plastic extrusion industry with the accumulated experiences and advanced technologies from continuous R&D activities.
MEK gauging systems and others are used as reliable quality control system for the various applications manufacturing the materials
for IT, display, secondary battery, metal foils, glass and steel as well as the conventional packing.

* Representative products: Online thickness gauge, Surface Inspection System, WEB cleaner, Pinning system.



History

지속적인 R&D개발로 수출증가 해외 지사 설립 (유럽 및 미국 법인 설립) Increased export, with sustainable Research & Development

- 2023 **MEK 창립 30주년 (Since 1993)**
MEK 미국 법인 (MEK GAUGING AMERICA LLC.) 설립
- 2022 MEK Europe 지사 설립
한국전지산업발전 유공자 기획재정부 장관상 수상
- 2021 모범납세자 부총리 표창 / Deputy Prime Minister's Commendation of exemplary taxpayers
- 2020 2차전지 Roll Press 공정 Line 측정기술의 World Best로써 차별화 된 공정제어 monitoring 기술의 Global 선도
With the world best measuring technology for the Roll press of secondary battery, leading the global monitoring technology with.
MEK Inc. 백만 불 수출의 탑 수상 / Acquire Million Export Tower Award
- 2019 **MEK 제 2공장 준공 / Construction of MEK 2nd Factory**
- 2018 MEK Gauging System CE / UL 인증 획득 / Obtained CE & UL certification for MEK Gauging System
- 2017 MEK Gauging System 디자인등록, 비접촉식 웹클리너 특허 취득
Registered design of MEK Gauging System, patented non-contact web cleaner
- 2016 **주식회사 엠이케이 법인설립 / Established corporation, MEK Inc.**
- 2014 비접촉식 웹클리너 개발 / Non-contact Web cleaner Developed
- 2011 수출유망중소기업지정 피닝 시스템 관련 특허 2종 취득
Appoint to "Competitive exporting company", Register 2 Patents for Pinning System
- 2009 **MEK 부설 연구소 설립 MEK 연구동 증축 / Founded MEK Research & Development Center**
- 2008 일본 대리점 계약 대만 대리점 계약 / Register the patent for MEK Pinning system
- 2007 맥엔지니어링 두께 측정기 Fine-Scan 발표(일본 플라스틱 협회)
Export MEK Gauging System(Fine-Scan) to Japan
- 2006 피닝시스템 관련 특허 등록 / Register the patent for MEK Pinning system

글로벌 기업으로의 도약, 해외시장 진출 본격화 A new leap towards becoming global company

- 2004 **본사 이전 (인천 부평국가공단 소재 / 사업자등록번호 변경) / Move Head Quarter to Incheon**
MEK Eng. 백만 불 수출의 탑 수상 / Acquire Million Export Tower Award
- 2003 인도 에이전트 계약 / Agency agreement for Indian market
ISO 9001:2000 인증 획득, 피닝시스템 수출 확대: 아시아, 유럽, 남미
Acquire ISO 9001:2000 certificate, Expand the exports of Pinning system to Asia, Europe and South America
- 2001 한국 최초 온라인 두께 측정기 수출 / Export Online gauging system

MEK의 탄생, 성장과 도약 Birth of MEK, Growth and Development

- 1999 R2R 공정을 위한 In-line 측정 시스템의 국내 최초 개발, 측정기술 국산화 및 국내 Film 산업 현대화에 기여
First Development for the in-line gauging system for R2R process in Korea leads to the localization of measuring technology and the contribution to the modernization for the domestic film industry
- 1995 IGBT형 코로나처리기 세계 최초 개발 및 판매 / IGBT type corona treater
- 1994 미국 EGS(Eurotherm Gauging Systems Inc.) Thickness Gauge
한국 독점 대리점 계약
Agency agreement with EGS(Eurotherm Gauging Systems Inc.) of US
피닝 시스템 개발, 일본/독일 수출
Pinning system, export to Japan/Germany
- 1993 **맥엔지니어링 코퍼레이션 창립**
Established MEK Engineering Corporation

R&D Center



연구 분야

R&D area

- 신제품 개발 - New systems
- AI제어시스템 개발 - AI Control system
- 신규 센서 개발 - New sensor

연구 조직

Organization

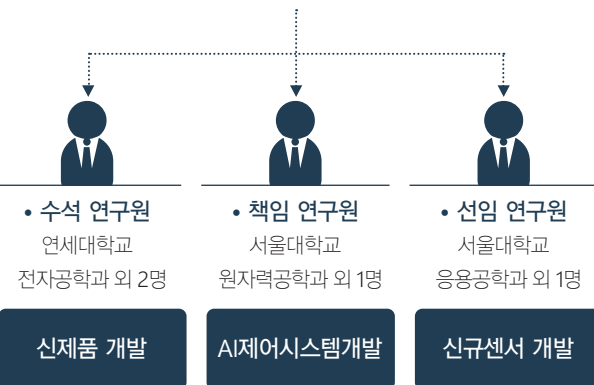


- 연구소장 (외 7명)
포항공과대학교 기계공학과

연구 설비

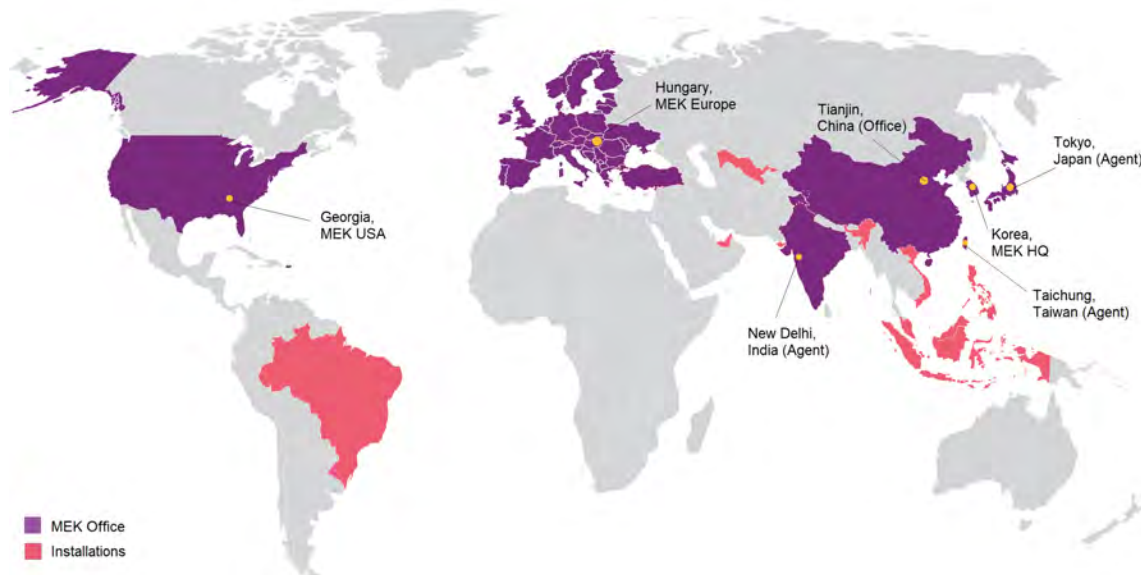
Facility

- Sheet 시험 제조 Line - Sheet test line
- On-Line 시험 측정 Line 1 - MEK Pilot Line 1
- On-Line 시험 측정 Line 2 - MEK Pilot Line 2
- Web Cleaner 시험 설비 (Clean Room) - Web cleaner in clean booth
- Pinning Demo Line - Pinning system
- Web 검사 Test 설비 - Web inspection system



Reference

일본, 미주, 유럽, 중국 등 26개국 1,000대 이상 System 제공



Partner - Major Customers (Region)

Battery

HYUNDAI MOTOR COMPANY.
SDI
LGES
SKON
PNT
CIS
SKNX
LOTTE EM
JPNG
Nan Ya Plastics Corporation
Chang Chun Petrochemical

KOREA

Samsung Electromechanics
Samsung Electronics
Samsung R&D
LG Chemical
LG Electronics
LG Hausys
LG R&D (Techcenter)
SKC
SKC R&D
Hanhwa Advanced Materials
Hanhwa R&D

Doosan Electronics
Doosan Heavy Industries
Doosan R&D
Hankook Tire
Lotte Chemical
Hyosung Chemical
Hwaseung Chemical
Kolon Industries
Bayer Material Science
Albany International
Toray Advanced Materials
Hankook Tire

CHINA

Lucky Science & Technology Industry
Genzon Novel Materials
Jiangdong Electronic Materials
Zhangjigang Kangde Xin Optronics
Jiangsu Xingye Juhua Co Ltd
Shuye Environmental Technology
Shaoxing Xiangyu Green Packing
Bangkai Plastics
Yingkou Kanghui Petrochemical
SKC
Hyosung
Hankook Tire

JAPAN

Japan Steel Works
Hitachi Zosen
Sumitomo Modern
Toshiba Machine
Mitsubishi Heavy Industries
Sekisui Chemical
Futamura Chemical
Gunze
Bando Chemical Industries
SANBO
SHEEDOM

SOUTH-EAST ASIA

Asian Pulp & Paper
Trias Sentosa
A. J. Plast
Polyplex
Samsung Electromechanics
Hankook Tire
ILJIN Materials
MIDDLE EAST
UFLEX Limited
JBF RAK LLC

MIDDLE EAST

UFLEX Limited
JBF RAK LLC

AMERICA

TERPHANE
LG Hausys
SKC
Hankook Tire

INDIA

UFLEX Limited
JINDAL Polyester
Polyplex
ESTER Industries
SRF Limited

TAIWAN

Shinkong Materials Technology
Nan Ya Plastics Corporation
Chang Chun Petrochemical
TEXYear

EUROPE

Lindauer Dornier
Brueckner Machinebau
Synthesia (Aliachem AS)
Starlinger
Circuit Foil Luxembourg
Hankook Tire

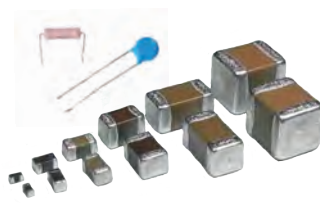
MEK On-Line Thickness Gauging System

적용분야 Application



전지 / Batteries

동박(Copper Foil)
분리막(Separator)
전극 코팅(Electrode Coating)



전자소재 / Electronic materials

가전 부품(Home Appliances Parts)
MLCC, EMC
코팅(Coating)



디스플레이 / Display

확산판(Diffusion Plate)
도광판(Light guide plate)
코팅 필름(Coated Film)
보호필름(Protective Film)



금속 / Metal

금속 시트 (Metallic Sheet)
메탈 시트 (Metal Sheet)



포장용 / Packaging

식품포장(Food Packaging)
연포장(Flexible Packaging)
PE, PP, PET, Etc.



기타 / Etc.

태양광(Solar Light), 발포 시트(Foam Sheet)
PVC필름/시트(PVC Film/Sheet)
EVA시트 (EVA Sheet), 타이어(Tire)
내장재(Interior material)
인조가죽(Artificial leather)

WEB 공정의 필수, 온라인 두께 측정기 On-Line Thickness Gauge for WEB manufacturing Process

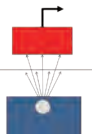
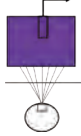
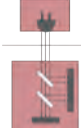
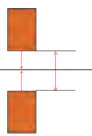
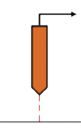
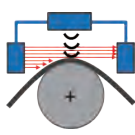
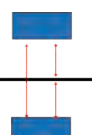
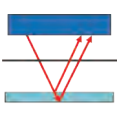
MEK 온라인 두께측정기는 웹(WEB)형상의 제품 두께와 무게를 측정하며 두께, 무게, 밀도를 해석하는 장비입니다. 제품 생산 중에 전폭의 두께를 실시간 측정 및 제어함으로써 품질향상과 공정 최적화 및 원가절감에 도움을 주는 세계 최고 수준의 제품입니다.

MEK on-line gauging system measures the thickness of web during production and control to get best thickness profiles with measured data. That enables to enhance the product quality and optimize the production process.

시스템 적용 효과 Advantage of using on-line thickness gauging system

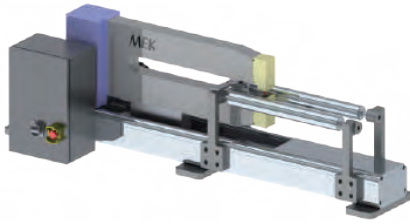
- 두께, 무게, 밀도 외 편차 관리로 인한 품질 향상 - Improved product quality
- 측정 데이터 보존을 통한 제품 관리 - Product management
- 공정관리로 시스템 최적화 - System optimization
- 원료, 원가 절감 - Reduced raw material consumption

센서종류 | Type of Sensor

Sensor	Description	측정범위 (Range)		
		Film	Sheet	Plate
X-RAY	 저 에너지 X선을 사용하는 투과식 측정법으로 측정체의 성분, 두께, 무게의 분석이 가능. As a measurement technique using X-ray in transmission mode, the sensor analyzes composition, thickness, and weight of sample.	1gsm ~ 350gsm	1gsm ~ 4,000gsm	
β -RAY	 베타 붕괴에 따른 전자 입자선을 사용하는 투과식 측정법으로 측정체의 두께, 무게의 분석이 가능. As a measurement technique using Beta decay electron particle in transmission mode, the sensor analyzes thickness and weight of sample.	1gsm ~ 175gsm 3gsm ~ 1,300gsm	76gsm ~ 7,600gsm	
IR	 적외선 대역을 이용한 투과식 및 반사식 측정법으로 측정체의 성분, 두께, 무게의 분석이 가능. As a measurement technique using Infrared either in transmission or reflection mode, the sensor analyzes composition, thickness and weight of sample.	0gsm ~ 200gsm		
IR (Interference)	 2개 영역의 파장을 사용하는 투과식 측정법으로 측정체의 성분, 두께, 무게의 분석이 가능. As a measurement technique using two waves in different wavelengths in transmission mode, the sensor analyzes composition, thickness, and weight of sample.	1gsm ~ 200gsm		
CONFOCAL Dual	 동축 변위 센서를 이중으로 사용한 측정법으로 두께의 직접 측정이 가능. As a measurement technique using confocal sensors in dual mode, the sensor directly measures thickness of sample.	1 μ m ~ 500 μ m		
CONFOCAL Reflective	 백색광을 사용하는 반사식 측정법으로 측정체의 두께 분석이 가능. As a measurement technique using white-light in reflection mode, the sensor analyzes thickness of sample.	0 μ m ~ 100 μ m		
LASER Roll	 선형 array Laser 센서를 보상 센서와 연동하여 Roll 상에서 측정하는 기법으로 측정체의 두께의 직접 측정이 가능. As a measurement technique using back-up roller, Line-array Laser sensor and auxiliary compensation sensor, the sensor directly measures thickness of sample.	200 μ m ~ 5,000 μ m		
LASER Dual	 반사식 Laser 센서를 이중으로 사용한 측정법으로 측정체의 두께의 직접 측정이 가능. As a measurement technique using reflective Laser sensors in dual mode, the sensor directly measures thickness of sample.	0.5mm ~ 10mm	5mm ~ 60mm	
LASER Reflective	 반사식 레이저 센서를 이용하여 투명 소재만 측정이 가능하며 표면 반사와 배면 반사 간의 거리 차를 두께로 환산. Reflective LASER sensor can only measure transparent material by converting distance value between surface and rear reflectance to thickness value.	50 μ m ~ 3,000 μ m		

스캐너 프레임 | Scanner Frame

- C-Type
(~1,400mm)



- 전지용
(~750mm)

특허출원
Patent pending



- 광폭전지용
(~1,500mm)

특허출원
Patent pending



- Compact O-Type
(2,500mm)



- General Roll Type
(~4,000mm)



- 정밀형
(~800mm)



- Medium O-Type
(2,500-5,500mm)



- Large O-Type
(5,500-9,000mm)



- Extra Large C-Type
(8,500-12,000mm)



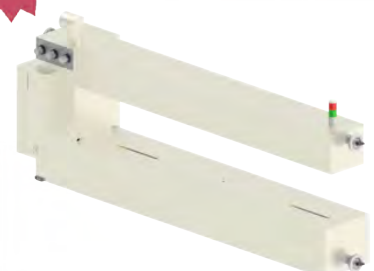
- Laser S-Type



- X-Ray Type
(~3,000mm)



- X-Ray Type
(~3,000mm)



프로그램 | Software

사용자의 이해와 최적의 공정 조건을 위해 다양한 방식으로 해석한 데이터를 보여주며 SQC등 각종 통계를 제공.
Display the measurements such as SQC for easy monitoring which ensures optimal production condition.

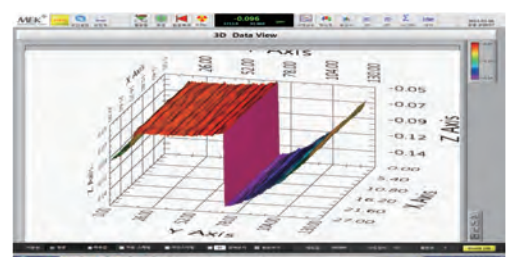
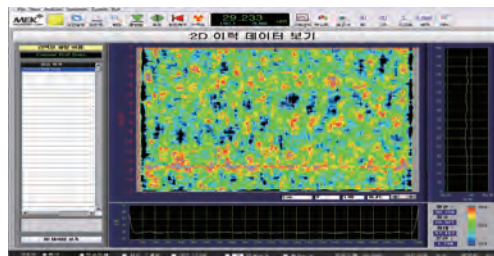
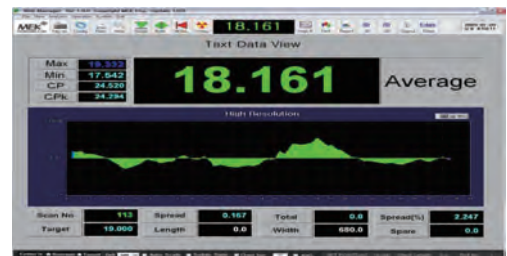
MEK 프로그램 특징

MEK Software
Features

- 사용자 친화적 구성 - 직관적 인터페이스 - Easy access to measured and saved data
- 언어 선택가능 한, 영, 중, 일 - Software in Korean, English, Chinese, Japanese
- User HMI와 측정 데이터 통신(선택사양) - Data communication with user HMI
- 사용자 요청에 맞춤 제작 가능 - Customized software

기본화면

Main Screen



APC | Automatic Profile Control

측정기와 연동하여 측정된 데이터를 분석, 일정한 공정을 유지하도록 Actuator을 실시간으로 제어하는 장치.

A system that enables real-time control of actuator to maintain process stability based on analysis of measurement data.

APC특징

Features of APC

- 최적화된 두께 제어 알고리즘 - Optimized thickness control algorithm
- 신속한 두께 제어 - Fast thickness control
- Auto Mapping: 센터볼트 기준 - Auto-mapping: From center bolt
- 매끈한 Mill Roll 외관 - Attractive mill roll appearance
- 수 Micron이내의 Control Gain 자동 조정 - Automatic adjustment for control gain of a few microns
- 다이볼트 온도 피드백 제어 (선택사양) - Feedback control for the temperature of die bolt (Optional)
- MD (Machine Direction) 제어 가능 (선택사양)

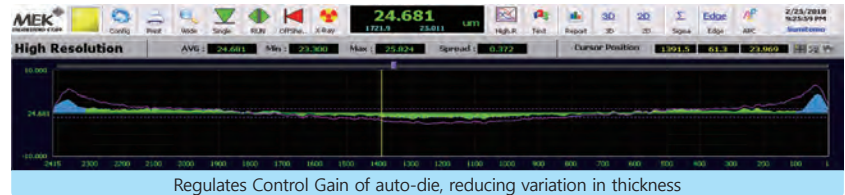
APC적용 후 프로파일 변화

Thickness profiles
after APC control

Before



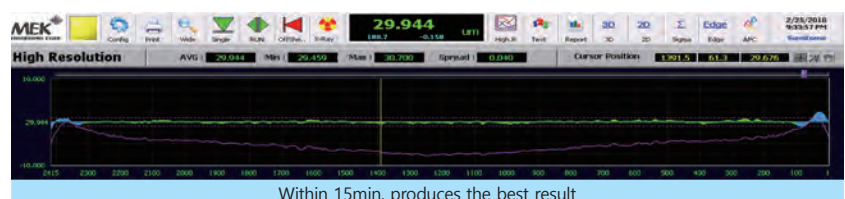
After 5min



After 10min



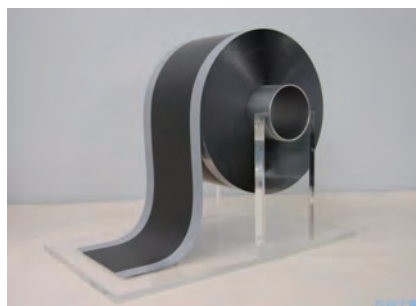
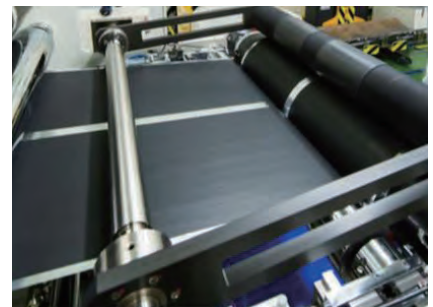
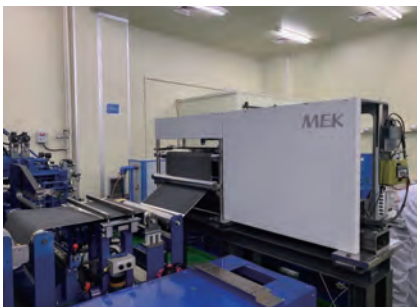
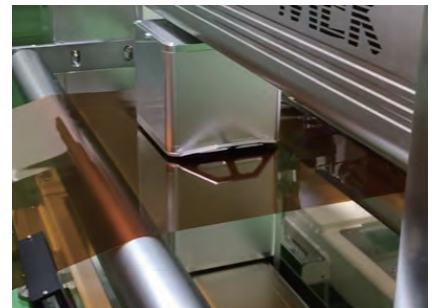
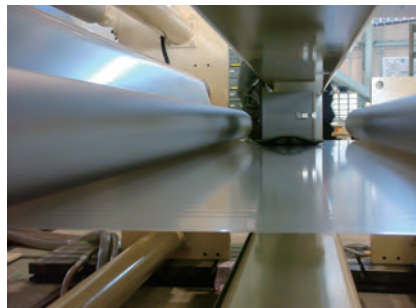
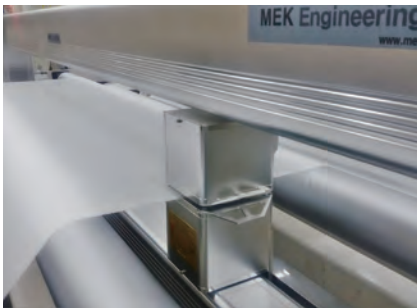
After 15min



Secondary Battery

적용 공정 | Application

1. 분리막 제조 - LiBS 2. 분리막 코팅 - Coating for separator film 3. 동박 제조 - Copper foil 4. 알루미늄 박 제조 - Al foil
5. 극판 코팅 - Electrode coating 6. Roll Press - Roll press 7. 파우치 제조 - Pouch

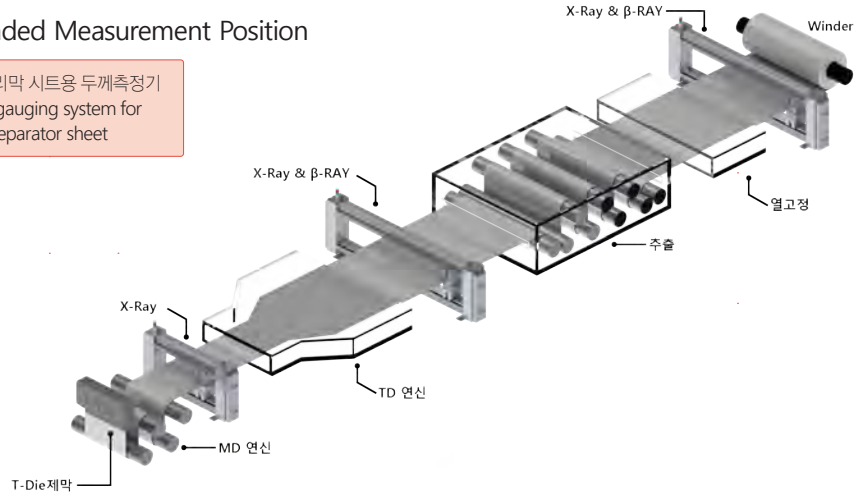


1. 분리막 제조 | LiBS

Battery BIAx
Battery
Separator Film
Manufacturing
Process

• Recommended Measurement Position

MEK 분리막 시트용 두께측정기
MEK gauging system for
separator sheet



분리막 두께 측정 및 제어

Separator Thickness
Measurement and
Control

• 분리막 - Separator Film



분리막 제조라인의 두께 측정 및 자동제어
Thickness measurement and automatic control of separator line.

• 배터리 코팅 - Coated Battery Film

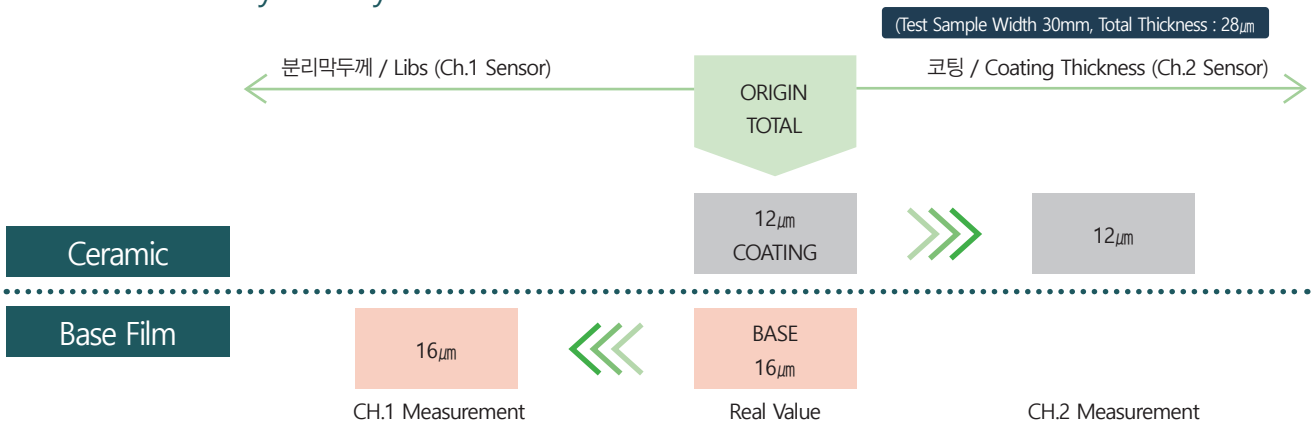


구간별 두께 측정 및 제어
베이스 두께를 별도로 측정 후 코팅부 두께 관리 및 제어 가능
Measurement and control of thickness by section.
Control of coating thickness and control after separately
measuring base thickness.

2. 분리막 코팅 | Coating for separator film

2차 전지 Base(분리막) 및 Coating(ceramic) 두께동시, 분리측정

Individual measurements for coating and substrate layers of ceramic coated separator film for secondary battery



3. 동박, AL박 제조 | Copper foil, Al foil

MEK 온라인두께 측정기

MEK On-Line
Thickness Gauging
System

측정 범위 Measuring Range	센서 Sensor	재현성 Repeatability	적용 공정 Application Process	품질관리 Quality Management
~80 μm (Cu foil, 15kV) ~200 μm (Cu foil, 40kV) ~150 μm (Al foil, 15kV) ~12,000 μm (Al foil, 30kV)	X-ray	$\pm 0.0002 \mu\text{m}$ (Cu)	제박 슬리터 오프라인 (ED / Slitter / Offline)	두께관리한도 초과시 Alarm발생 Alarm occurrence if thickness management limit is exceeded Roll data 저장 / Roll data storage

동박 두께 측정

Copper Foil

Overlapped graph of 10 times measurements

[MEK 두께 측정 데이터 / MEK's Performance]

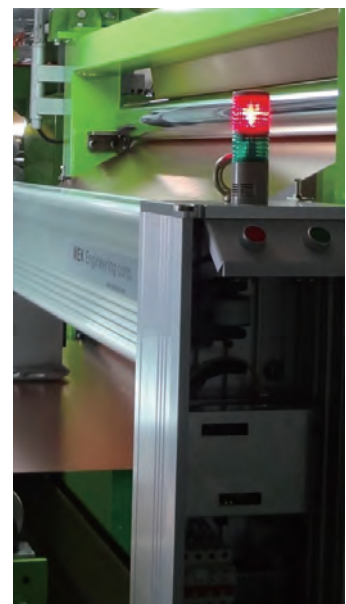
10번의 측정 시 측정 위치가 거의 동일. 그래프 내 10개의 선이 겹쳐져 하나의 뚜렷한 선을 보이며 센서의 재현성이 매우 높음.

The measurement position is almost the same in 10 measurements. 10 lines in the graph overlap to show one distinct line, the sensor is very reproducible.



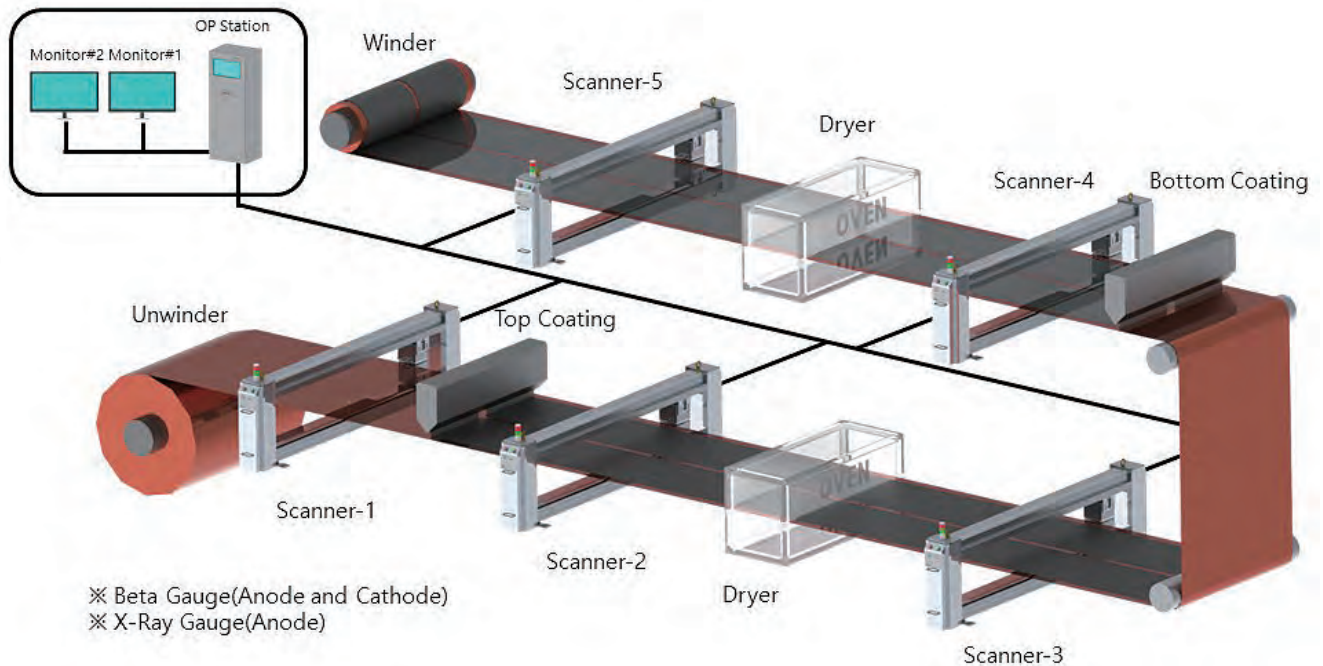
Cu/Al Foil MEK 두께 측정기 적용

in-Line Application



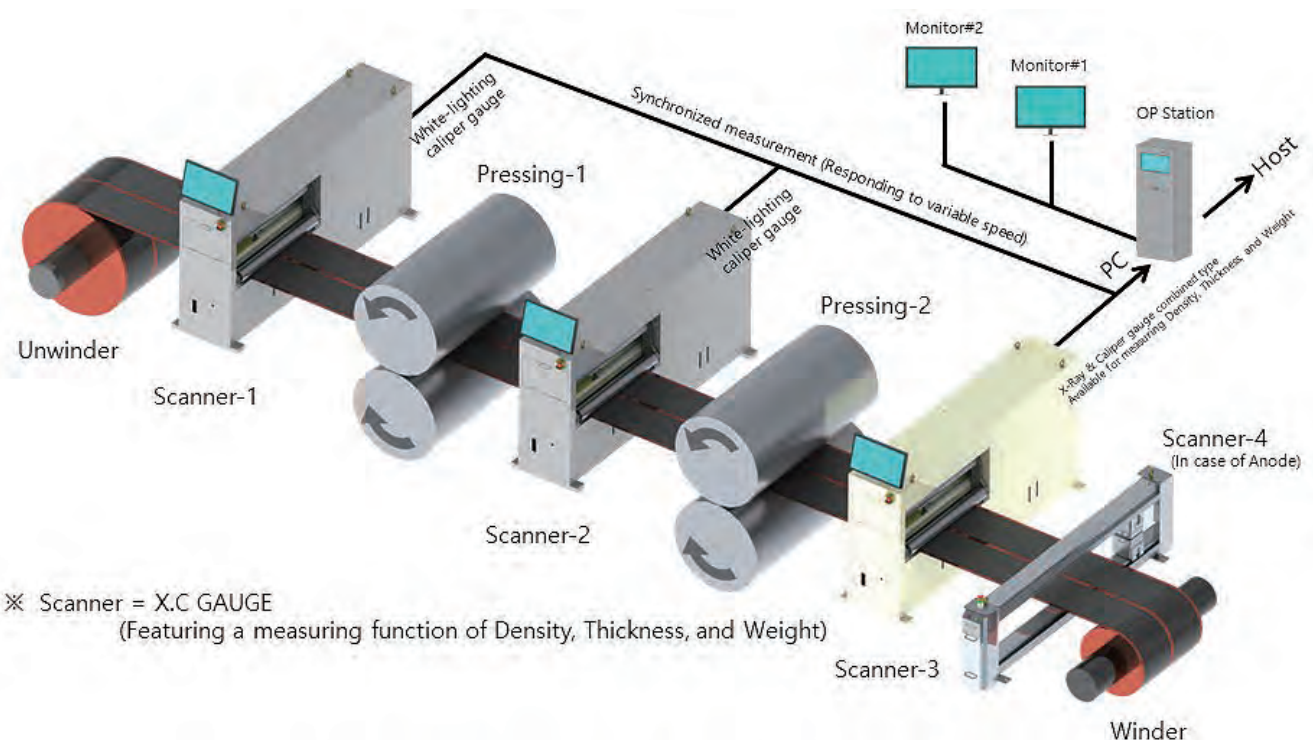
4. 2차전지 양극/음극 코팅 공정:WEB GAUGE 적용사례

Secondary battery anode/cathode coating process



5. 2차전지 양극/음극 프레스 공정:WEB GAUGE 적용사례

Secondary battery anode/cathode press process



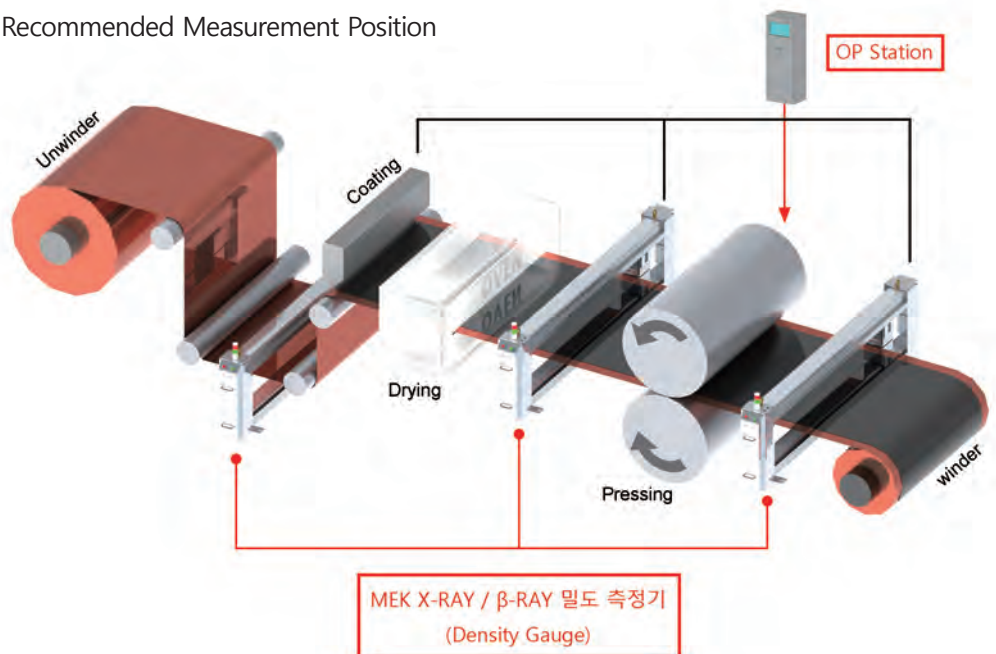
6. X-선 & β -선을 이용한 극판의 밀도/두께 측정

Measurement of the density / thickness for the anode and cathode with X-ray & β -ray.

코팅 및 롤프레스/
카렌다 공정을 위한
밀도 /두께 측정 및
제어

Measurement and
control of the density
and thickness for
coating, roll press/
calender process

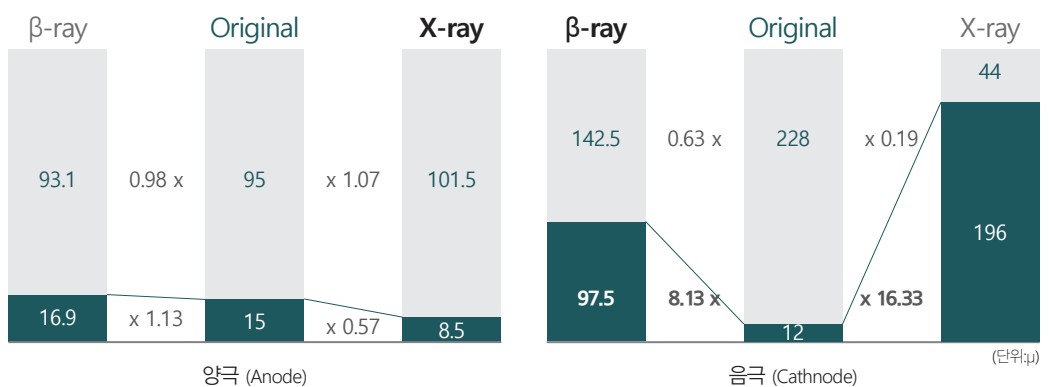
- Recommended Measurement Position



흡수율 변화

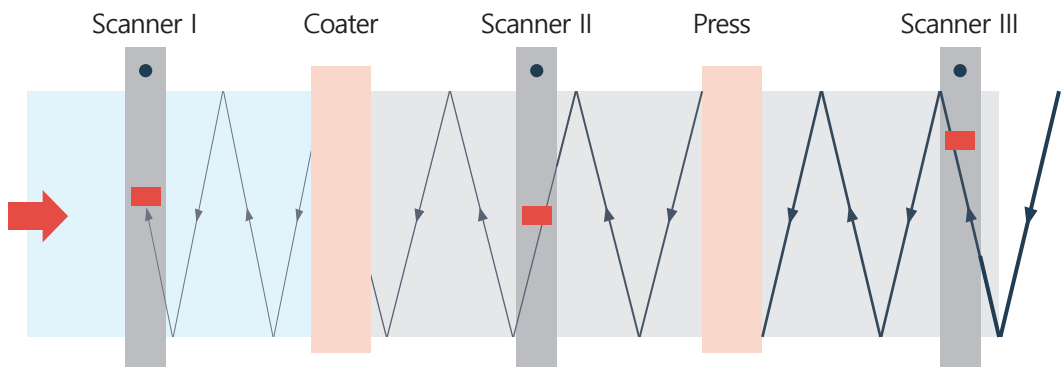
Absorption ratio for
the anode and cathode

- MEK Quality Control Online System / Thickness Gauge



동기화 측정

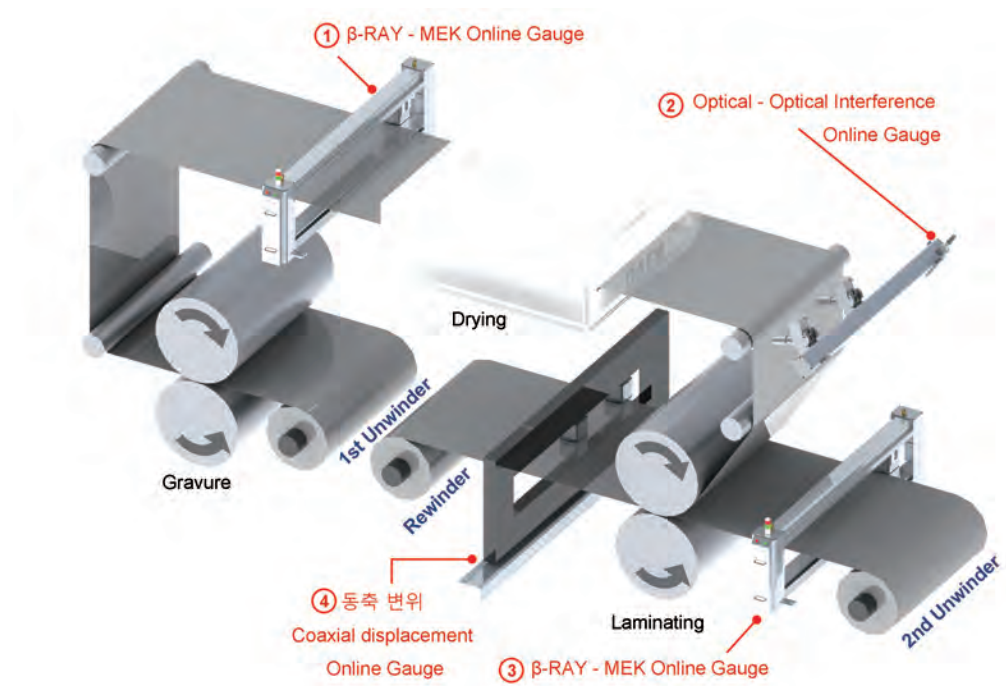
Synchronized
Measurement



- 코팅, 프레스 후 같은 지점을 측정
Measurements on the same spot after coating and pressing processes
- 층별 두께, 밀도, 해석 가능
Analyzing thickness, density of each layer.

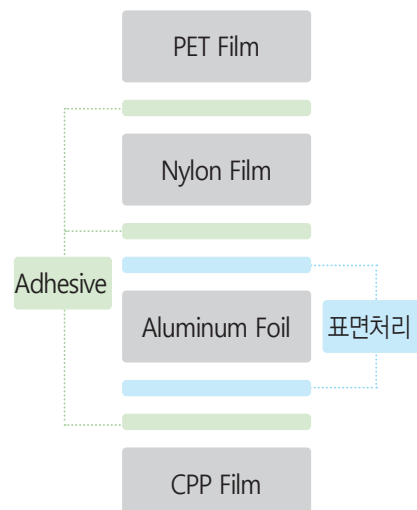
7. 파우치 제조 | Pouch

Battery Pouch Manufacturing Process



• Recommended Measurement Position

공정 & Sheet Process sequence & sheet composition



8. Roll Press 후에 사용되고 있는 동축 변위 센서 측정 원리

Measuring principle of confocal sensor used after Roll Press

Roll Press 후에 사용되고 있는 동축 변위 센서 측정 원리

Measuring principle
of confocal sensor
used after roll press

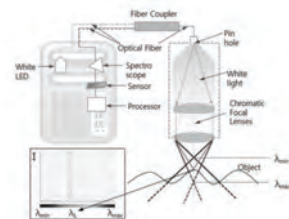
백색광을 색상별 (파장별)로 집광 위치를 변화시켜 조사하고 초점을 맞춘 파장광만 반사광으로 수광 한 뒤 해당 색 정보를 통해 변위를 측정하는 원리이며 MEK 공초점 Dual 센서는 진동과 Tension 변화의 영향을 극소시켜 Roll Press 공정 현존 최고의 두께측정기로 평가 받고 있음.

MEK는 2017년 공초점 센서를 이용 Roll Press 공정에 세계최초로 적용한 원조 회사임.

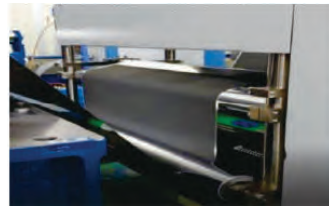
White light is separated spectrally by colors (by wavelength), and only focused light is reflected. By analyzing it thickness is measured. In MEK confocal sensor, the influences of by vibration and irregular tension are minimized.

That resulted in state-of-art thickness gauging system for roll press.

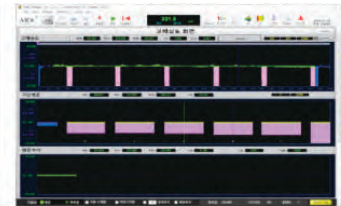
MEK is the world's first aid company to apply to the Roll Press process using confocal sensor. sensors in 2017.



sensor



Roll Press



Measuring Profile

9. 신제품

① 2차 전지 ♡ 용 I 4 r i n 6 X e A J A

설명 : 2차전지 극판에 최적화된 제품으로 두께 중량 동시 계측하여 밀도까지 해석 가능

▶ 전용 : X-ray or β-ray + 공초점 센서

※ ㄴ D 접촉식 추가 가능



특허 등록번호
10-2400495

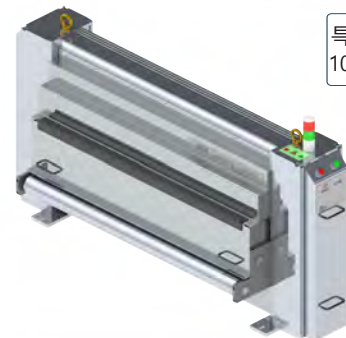


② 전고체 Battery ♡ 용 I 중량 (두께) 측정기

설명 : 전고체 배터리에 최적화된 제품으로 두께 중량 동시 계측하여 밀도까지 해석 가능 // x-ray 방호용 장비로 신고만으로 사용 가능

▶ 전용 : X-ray + 공초점 센서

※ 방사선 관리 불필요



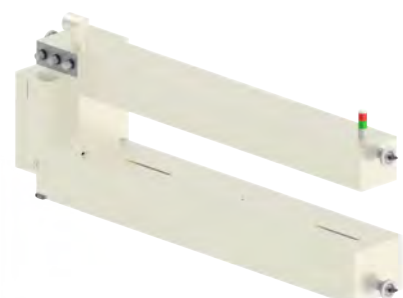
특허 출원번호
10-2022-0081423

③ 금속 sheet용 I 고에너지 X-ray Gauge

설명 : 고출력 3 point 7 ㄹ ㄴ ㄴ 또는 ㄴ 스캔 가능 장비

▶ 전용 : AL, STEEL, CUPPER SHEET 등

※ 허가 승인 필요



1차 전지 분리막 두께 측정 및 제어

Thickness Measuring & Control for the Separator Film of Primary Battery

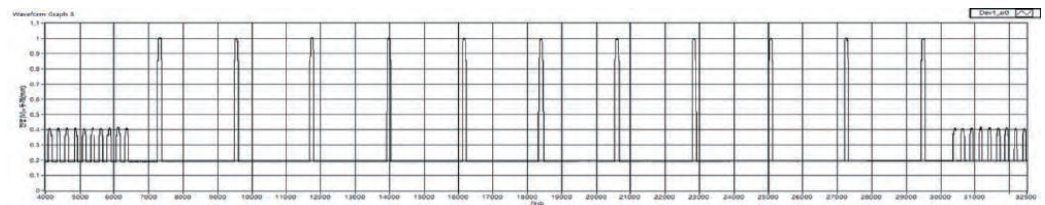
1차 전지 분리막

Separator Film of
Primary Battery

- 1차 전지 분리막 표면 확대 이미지 - BASE가 되는 표면위 RIB이 일정한 간격을 이루고 있는 형태
Enlarged view of the separate film for primary battert Constant distances of RIBs on the surface of BASE



- MEK 두께측정기 1차 전지 분리막 샘플 두께측정 데이터
Measurement for the separate film of primary battery by MEK gauging system



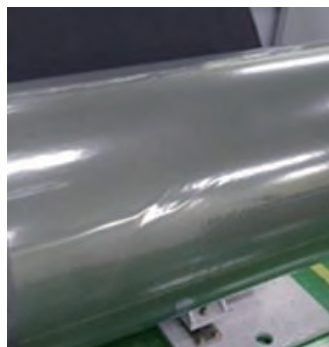
구분	1	2	3	4	5	6	7	8	9	10	11
RIB	1006	1001	1008	1004	1003	997	1001	1000	1001	1003	1000
BASE	195	190	191	188	191	193	195	194	196	197	X

연료전지

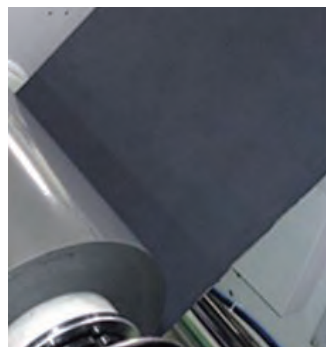
Fuel Cell

수소연료전지 두께/ 중량 측정

Measurement of
thickness/weight for the
film and sheet used in
fuel cell



Carbon Paper



GDL



현장적용사진

카본페이퍼 및 카본페이퍼를 이용한 GDL과 흑연펠트, MGP의 중량 및 두께 실시간 제어할 수 있음
Measure and control the weight and thickness of the carbon paper and GDL using it, carbon pelt and MGP in real time.

검사기 | On-line Surface Inspection System

Configuration

WebScan System - Including 5 major composition:

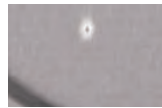
- **Optical** : precision optical equipment of Germany made CCD and lens with mounting bracket.
- **Illumination** : High frequency, high, high brightness LED light source.
- **Computer** : IPC, VNC module.
- **Electrical Control** : Including precise constant voltage regulator, power control, I/O control, alarm signal, speaker, holder, marker (Option).
- **Software** : WebScan - According to different requirements, the systems are able to add dust removal, interference against, colling system and other equipments.

Common Defects

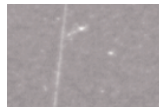
Optical filmsurface: Diffuser film, Brightness enhanced film



Dark dot



Bubble

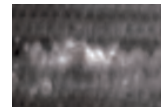


Scratch

Electronic products:Prepreg, FPC, CCL, LCD panel.



Fisheye



Disruption

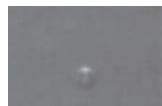


Dot defect

Tape inspection:Protect film, Tapes, Seperate paper, Label



Oil pollutant



Bubble



Contaminant

Textile industry:Paper, NCR, paper board, nonwaven...



Pinhole

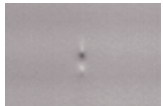


Crack



Insect

Plastic Film:BOPP, PP, PE, PET, Plastic Films and sheets...



Fish eye



Foreign matter



pinhole

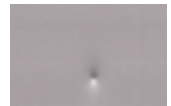
Metal surface inspection: brass foil, aluminum foil, tin sheet



Pit



Crease



Dents

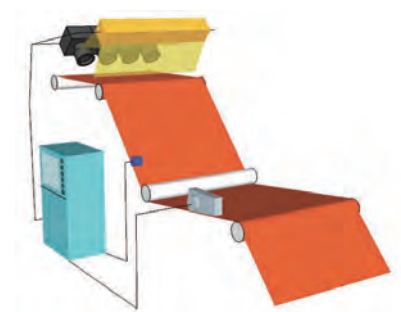
Application

Inspect defects:

($\varphi 0.05\text{mm}$ above, or line speed less than 1,000m/min)

1. Surface Defects : scratches, creases, pits and dents.
2. Pinholes : Punctures pinholes.
3. Bubbles : Air bubbles trapped in clear materials of punctured or imbedded ones.
4. Spot, impurities,foreign matters, i.e. dirt, inclusion.
5. Foreign Matters in the form of insect fragments.
6. Pitted edges, creases, and lateral/vertical line disruption.
7. Uneven composition, oxidize and color change(stain) etc.

Others : Contrast exceeding ± 30 in gray scale compare with mass material.

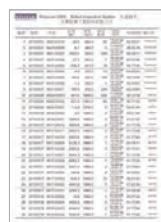


Basic Principle

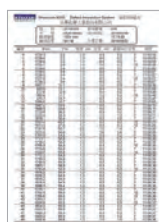
Software & Reports

According to different demand, we provide various reports for export.

• Available Reporta Formats



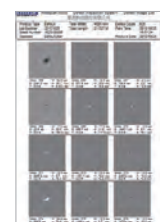
Production List



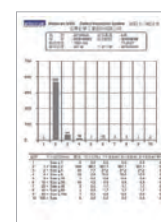
Defect Data List



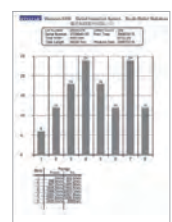
XY Map



Defect Image List



Defect Size Statistics



Defect Aspect Ratio Statistics



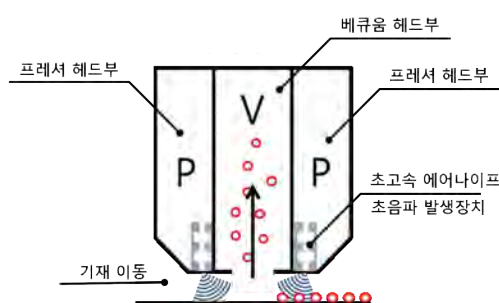
특허 2014-05-22

웹크리너 Non-Contact Web Cleaner

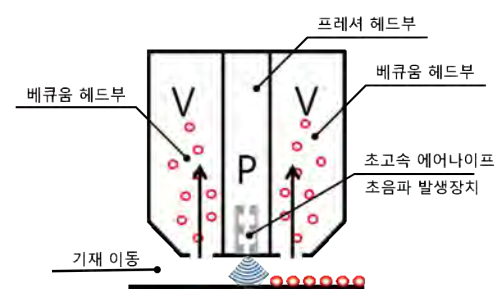
시스템 개요 Systems

생산 공정상(On-line)에서 비 접촉식으로 WEB 형상(Sheet/film)의 제품 표면의 이물을 제거하는 장치.
On-line, non-contact cleaner removes the foreign particles on the web surface.

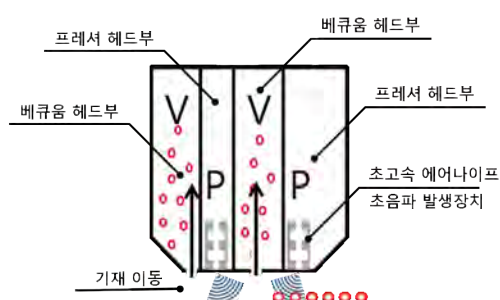
작동원리 How it works



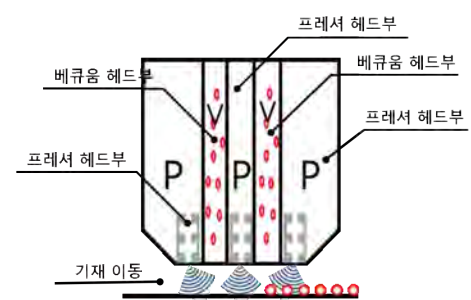
■ PVP



■ VPV



■ VPVP



■ PVPVP

** (P : Pressure / V : Vacuum)

웹크리너 | Non-Contact Web Cleaner

Vacuum부와 pressure부로 나누어진 헤드에서 공기가 나가고 먼지를 흡입하는 과정에서 생산 공정 중 제품 표면이 깨끗해지는 효과를 볼 수 있음.

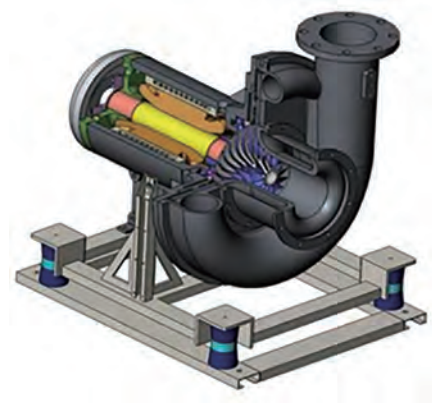
제품특징

Features

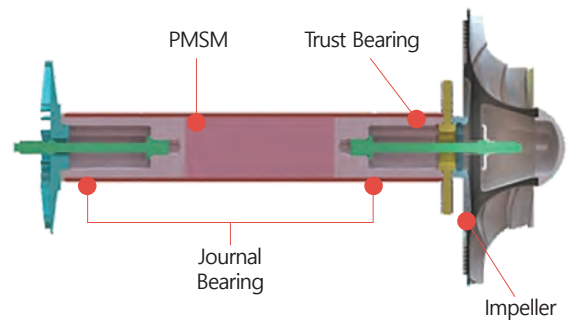
- 강력한 초음파 발생: 138dB 이상.
Ultrasonic wave generation: 138dB or above.
- 프로그램에 의한 풍압 및 풍량 자동제어 가능.
Automatic control of pressure & volume.
- 헤드 열변형 방지 설계: 제품 손상 방지.
Design for no thermal deformation: Prevent the scratch on the product due to irregular thermal expansion.
- 모터 및 임펠러 동축 구조, 공중부양 구동으로 그리스-Free구조이며, 초고속 에너지 혁신성 제품. (소음85dB이하, Rpm: 15000~50000)
Motor & Impeller co-axial structure with air foil bearing employed. Energy saved & Noise reduced. (Below 80dB, Rpm: 15000~50000)
- 지지대 및 공기 배관 일체형으로 호스가 불필요하므로 심플하고 스마트한 설계로 슬리터와 같은 협소한 설치 공간 대응 가능.
Simple and smart design of integrated support frame and air duct need no air hose. That overcome space constraint at slitte.
- 유지보수에 유리: 제품 내 LM가이드, 블로워 베어링, 벨트 등이 없어 유지비용 감소와 Duct 내부 오염 원인 제거.
Maintenance-easy: There are no parts that require frequent maintenance such as blower bearing and belt.

MEK 터보 블로워

Turbo Blower



직접 연결 방식
Direct Connection



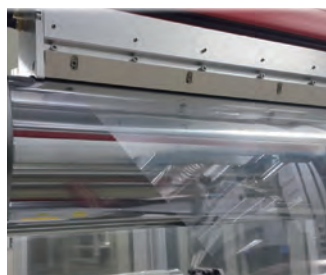
구조
Structure

적용 사례

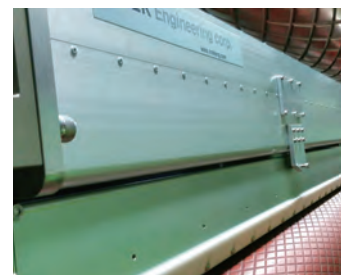
Application



테이크-업 / Take - up



코팅 프로세스 / Coating Process



슬리터 / Slitter



피닝 시스템 Pinning System

시스템 개요 Systems

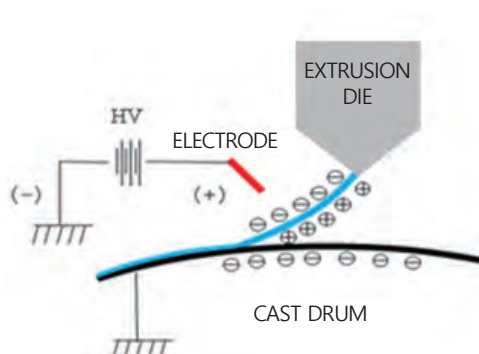
플라스틱 Film / sheet 압출 공정에 사용되는 장치로, 용융 수지의 냉각 Roll 에 대한 밀착을 비접촉, 전기적 방식으로 제공하므로 Nip-roll을 이용한 Calendar 방식 혹은 Air knife를 이용한 방식에 비하여 밀착 성능이 균일하고 우수.

Non-contact, electrical method of applying molten polymer on to cooling roll used in plastic film/sheet extrusion process, provides uniform and superior performance compared with air knife or calendar method using nip-roll.

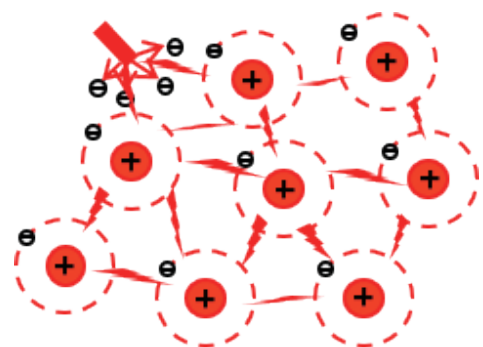
작동원리 How it works

고전압이 인가된 전극을 이용하여 용융 수지를 대전 시키면 정전기력에 의거한 자가 밀착성이 수지에 부여되고, 따라서 냉각 Roll 에 대한 균일하고 강한 밀착력을 나타냄.

Charging molten polymer using high-voltage electrode creates self-adhering property in the resin by static electric force and it therefore exhibits uniform and strong adhering force on to cooling roll.



분극현상
Polarization



방전에 의한 분자 이온화
Air Molecule Ionization By discharge

피닝 시스템 | Pinning System

적용분야

Applications

- PET 를 포함한 고분자 수지의 Film/sheet 압출 공정.
Film/sheet extrusion process of various resins including PET.

적용 효과

Advantages of Pinning System

- 에어나이프나 배큐움 박스와 달리 고압 전원의 인가로 발생하는 정전기력을 이용하므로 성능이 균일하며 밀착력이 우수.
High voltage is applied on the wire or band electrode.
- 표면성 및 투명도가 대폭 향상되며, 폭 방향 두께 프로파일 개선 및 변부 수율 개선으로 제품 폭 증가 및 TDO 파단 감소 등의 이점이 있음.
Electrical pinning show uniform and superior performance comparing with air knife and vacuum box. Enables clear and enhanced surface, better CD thickness profile, enlarged product width from less neck-in and less breakage of TDO.

SYSTEM 종류

Types of system

- PWire, Doble Wire, 4mm Band, 8mm Ω Type

피닝 전극

Pinning Electrode

Band			Wire
Size : 8mm(W) x 40 μ m(T) x 300(L)			Primary Electrode : Ø 0.1 / Ø 0.15 / Ø 0.2
Operation Tension : 8~25kg Maximum Tension : 75kg			
Precise Grinding On Edge Other Size			
T	L	W	Auxiliary Electrode : Ø 0.3 Tension : 0~4kg Type of Pinning System
40 μ m	100/300m	3/4/8/12mm	
50 μ m	90/25m	3/4/8mm	

시스템 구성

System Components



피닝시스템 (와이어 / 밴드)
Pinning System (Wire / Band)



오퍼레이팅 패널
Operating Panel

A photograph of a modern glass-fronted building, identified by the 'MEK' logo on its facade. The building is partially obscured by the dense, green foliage of several trees in the foreground. The sky is visible through the branches of the trees.

Your Technical Partner
for Web Application

(주)엠이케이 / 맥엔지니어링 코퍼레이션

MEK Inc. / MEK Engineering Corporation.

30, Bupyeong-daero 313 beon-gil, Bupyeong-gu, Incheon, KOREA

Tel. +82-32-516-1301 | Fax. +82-32-516-1069

Website. www.mekeng.com | E-mail. sales@mekeng.com

- China Office. +82-130-4324-4488
- Taiwan Agency. +886-2-8521-3680
- Japan Agency. +81-06-6446-9030